Assignment 6

Technical Analysis using R, Visualization Phase

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Course	BDA400 - Data Science Tools and Techniques
Assignment title	Assignment 6, Technical Analysis using R, Visualization Phase
Assignment repository	https://github.com/dipakghimire/BDA400_Assignment6

Project Description

This project is an interactive portfolio dashboard built with R Shiny. It downloads historical stock prices from Yahoo Finance using the quantmod package (with Stooq as a backup source if the download fails) and plots them with ggplot2 as a candlestick, line, or area chart. The user can change the stock symbol, date range, and time frame (daily, weekly, or monthly).

Moving averages, RSI, and MACD can each be switched on or off from the sidebar. A moving average crossover rule generates Buy, Sell, and Hold signals: a Buy is signalled when the short-term average (20) crosses above the long-term average (50), and a Sell when it crosses below. The signals are annotated on the chart bars. The dashboard also includes a signals log that can be downloaded as CSV and a portfolio comparison tab.

Files in the Repository

DipakGhimire_BDA400_A06.R - complete R Shiny application, organized by assignment step (Step 1 to Step 4) with comments.
DipakGhimire_BDA400_A06_CoverPage.pdf - this cover page.